Married and Never-Married Days: Marriage, Gender and the Full Time Budget in India

Manuscript anonymised for review

Abstract

Objective: This study asks which activities differ alongside married Indian women's higher unpaid work, and how this varies within couples.

Background: Studies usually examine one outcome at a time, though a day's activities must sum to 1,440 minutes.

Method: Using all 450,457 diaries of India's Time Use Survey 2024, the study compares 88,337 married and never-married women and 87,364 men aged 20 to 39 across nine activity divisions. Selection is addressed by age restriction, reweighting and bias adjustment; 106,858 linked couples give within-household comparisons.

Results: Married women reported 280 minutes more of unpaid domestic and care work than never-married women; for men the difference was 26 minutes more. Reweighted, it was 233; bias-adjusted, 264. Wives did 367 minutes more of unpaid work than their husbands, and their share of the couple's unpaid work was 91%. Co-residence with parents-in-law made no detectable difference.

Conclusion: Marital status divides women's days far more than men's.

Keywords: marriage, time use, unpaid work, gender, patrilocality, India

Introduction

Marriage is closely tied to how time is used. In diary studies from high-income countries, married women do more housework than women in other living arrangements, and the gender gap in housework is largest among the married (Shelton & John, 1996; South & Spitze, 1994). Men's housework, in contrast, falls or stays flat when they enter a union (Baxter et al., 2008; Gupta, 1999). Explanations range from specialisation within the household (Becker, 1985, 1991) and bargaining over the division of labour (Lundberg & Pollak, 1993; Manser & Brown, 1980) to the performance of gender in everyday interaction (Akerlof & Kranton, 2000; Brines, 1994; West & Zimmerman, 1987).

India offers a sharp setting for these questions. In the Time Use Survey 2024, 99% of women and 95% of men aged 35 to 39 have ever been married. Residence after marriage is usually with the husband's family (Dyson & Moore, 1983; Khalil & Mookerjee, 2019): 29% of married women aged 20 to 39 in the survey are daughters-in-law of the household head. Married women's employment is low and has not risen with education (Afridi et al., 2018; Chatterjee et al., 2018; Klasen & Pieters, 2015). For most Indian women, marriage is the transition that sets the terms of adult time use.

Most studies of marriage and time use take one outcome at a time. Separate literatures examine housework (Coltrane, 2000; Lachance-Grzela & Bouchard, 2010), sleep (Burgard & Ailshire, 2013; Maume et al., 2010), free time (Mattingly & Bianchi, 2003), exercise (Nomaguchi & Bianchi, 2004) and social ties (Sarkisian & Gerstel, 2016). A day is a fixed budget, however. Any increase in unpaid work must come out of paid work, learning, leisure, social life, self-care or sleep. Looking at one outcome at a time cannot tell which, and a set

of separate "marriage penalties" estimated on different samples and specifications need not add up. For India, Janiso et al. (2024) use the Time Use Survey 2019 to decompose the large gap between women and men in unpaid housework and care work; this paper instead compares married with never-married women and men, across the whole day, using the 2024 round. This paper estimates the components together. I decompose the married–never-married difference into all nine divisions of the International Classification of Activities for Time-Use Statistics 2016 (ICATUS 2016, United Nations Statistics Division, 2016). The divisions partition the day, so the components sum to zero by construction, and sub-rows for sleep, socialising, exercise and religious practice show where these commonly studied outcomes sit within the budget.

The paper makes three further contributions. First, it treats selection into marriage as a central problem rather than a caveat. Never-married women in their thirties are rare in India, and they are likely to differ from married women in schooling, work orientation and health (Goldman, 1993). I report the decomposition for women aged 20 to 26, where never-married women are least selected; reweight the never-married to the covariate distribution of the married (DiNardo et al., 1996); compute bias-adjusted coefficients and breakdown values in the spirit of Oster (2019); and trace how the gap changes with age as the never-married become more selected. Second, it uses household structure to examine patrilocality directly. Comparing daughters-in-law of the head with wives who head their own household (or whose husband does) separates two accounts of the joint family: one in which the mother-in-law shares domestic work, and one in which the junior wife carries the heaviest load (Kandiyoti, 1988). Third, it links spouses' diaries within households and examines the within-couple

division of unpaid work by the wife's employment, spouses' relative education and the spousal age gap. These comparisons hold the household constant and do not depend on the never-married comparison group.

Married women aged 20 to 39 report 279.7 minutes (SE 2.5) more unpaid domestic and care work per day than never-married women of the same age, education, caste, religion and state. The corresponding difference for men is 26.2 minutes (SE 0.9) more, and the gendered difference in a pooled model is 261.9 minutes (SE 2.6). For women, the largest offsetting reductions are in learning (119 minutes) and employment (incl. travel) (85 minutes). Married women slept 9 minutes less than never-married women. Among linked spouses, wives' unpaid work exceeds husbands' by 366.9 minutes (SE 1.1) a day, and husbands' paid work exceeds wives' by 336.4 minutes (SE 1.5).

Background

Marriage and the division of household labour

The finding that marriage widens the gender gap in housework is among the most replicated in family research. South & Spitze (1994) show that women in married-couple households spend more time on housework than women in other living arrangements. Panel studies find that men's housework falls when they marry or cohabit (Gupta, 1999), and that women's housework responds more than men's to union and parenthood transitions (Baxter et al., 2008; Sanchez & Thomson, 1997). Trend studies document a narrowing of husbands' and wives' housework since the 1960s, but no convergence to equality (Bianchi et al., 2000; Kan et

al., 2011; Sayer, 2005). Cross-national studies show that the gap is larger where macro-level gender inequality is greater and where policies support a male-breadwinner arrangement (Fuwa, 2004; Hook, 2006, 2010). Reviews of this literature agree on the main correlates (relative resources, time availability and gender ideology) but not on their relative weight (Coltrane, 2000; Davis & Greenstein, 2009; Lachance-Grzela & Bouchard, 2010; Shelton & John, 1996).

Two families of theory organise this work. In economic models, spouses specialise by comparative advantage (Becker, 1985, 1991), home production competes with market work and leisure (Becker, 1965; Gronau, 1977), and in bargaining models a spouse's earnings and outside options buy relief from housework (Chiappori, 1992; Lundberg & Pollak, 1993; Manser & Brown, 1980; McElroy & Horney, 1981); relative-resources theory makes a similar prediction (Blood & Wolfe, 1960), and feminist economists add norms and extra-household institutions (Agarwal, 1997; Sen, 1990).

The second family treats housework as a way of performing gender (West & Zimmerman, 1987). Economically dependent husbands have been found to do less housework, not more (Brines, 1994; Greenstein, 2000), wives' housework can rise again once they out-earn their husbands (Bertrand et al., 2015; Bittman et al., 2003), and Akerlof & Kranton (2000) formalise the idea as identity. The evidence is contested (Gupta, 2007; Killewald & Gough, 2010), and structural accounts locate gender in institutions as well as in couples (Ridgeway & Correll, 2004; Risman, 2004), one reason couple-level resource effects may be weak where institutional constraints are strong.

The time budget as a whole

Time-budget research has long argued that activities should be studied together because the day is fixed (Gershuny, 2000; Juster & Stafford, 1991; Robinson & Godbey, 1997). Aguiar & Hurst (2007) and Gimenez-Nadal & Sevilla (2012) use complete time budgets to trace substitution between market work, home production and leisure. Burda et al. (2013) show that in rich countries men and women do roughly the same total amount of paid and unpaid work, while in poorer countries women do more. Marriage could therefore raise women's total work, or it could shift work from the market to the home with little change in total work. Only a full accounting distinguishes the two.

Free time also differs in quality: women's leisure is more often fragmented or combined with care (Bittman & Wajcman, 2000; Mattingly & Bianchi, 2003). Marriage may reshape social life too, as a "greedy" institution (Coser, 1974); married Americans have fewer ties to kin, friends and neighbours than the never-married (Sarkisian & Gerstel, 2016).

Sleep, health behaviour and marriage

Sleep is the largest single use of time, and it is not fixed. Biddle & Hamermesh (1990) show that sleep responds to the value of time, and in the American Time Use Survey work time is the activity most strongly traded off against sleep (Basner et al., 2007). Gender matters for sleep as well. Women's sleep is more often interrupted by caregiving (Burgard & Ailshire, 2013), and qualitative studies describe a "fourth shift" of night-time care that mothers rather than fathers carry (Maume et al., 2010; Venn et al., 2008). Self-reported sleep problems follow both gender and socio-economic position (Arber et al., 2009). If marriage

adds morning and evening domestic tasks (cooking before others wake, cleaning up after others eat), sleep is a plausible margin of adjustment.

Marriage is also associated with health behaviour and health (Umberson, 1992; Waite, 1995), partly through selection of the healthy into marriage (Goldman, 1993), and with exercise time in gendered ways (Nomaguchi & Bianchi, 2004). Sleep, exercise and personal care fall in divisions 8 and 9, so their marriage gaps can be read against the gaps in work.

Marriage in India

Three features of Indian marriage bear on time use. The first is patrilocality. A woman usually moves into her husband's household at marriage, and in many regions marries outside her natal village (Dyson & Moore, 1983; Rahman & Rao, 2004). In a joint household, the new wife is junior to her mother-in-law. Kandiyoti (1988) describes a "patriarchal bargain" in which young brides accept subordination and heavy work in return for authority later in life, when they become mothers-in-law. Recent studies find that co-residence with in-laws is associated with less say over household decisions and less freedom of movement (Debnath, 2015; Khalil & Mookerjee, 2019), and that mothers-in-law restrict daughters-in-law's mobility and social contacts outside the household (Anukriti et al., 2020). Regional differences in women's autonomy are large and are not explained by religion alone (Jejeebhoy & Sathar, 2001). Joint residence could reduce a young wife's domestic load if senior women share the cooking and cleaning, or increase it if the junior wife does the heaviest work. Its effect is an empirical question that the household roster can address.

The second is the timing and form of marriage. Most marriages are arranged, though

women's say has grown (Allendorf & Pandian, 2016); age at marriage reflects community gender scripts (Desai & Andrist, 2010), labour-market opportunity (Jensen, 2012) and schooling (Field & Ambrus, 2008). Women who remain unmarried in their late twenties and thirties are therefore a selected group, the main threat to any married–never-married comparison.

The third is women's employment, which is low and fell for much of the 2000s and 2010s, especially among married rural women (Afridi et al., 2018; Fletcher et al., 2017; Mehrotra & Parida, 2017), with explanations ranging from rising incomes and labour demand to status and mobility norms (Dhanaraj & Mahambare, 2019; Eswaran et al., 2013; Field et al., 2021; Jayachandran, 2021; Klasen & Pieters, 2015). Women carry most unpaid and own-use work (Hirway & Jose, 2011). If marriage moves women out of paid work and into domestic work, the time budget will show it; if it adds domestic work on top, leisure, social time or sleep must absorb it.

Hypotheses

The theories above make overlapping predictions about the size of the marriage gap and distinct predictions about its sources and moderators. I state them as expectations that the data can contradict.

H1 (gendered reallocation). Marriage is associated with a larger increase in unpaid domestic and care work for women than for men. Specialisation, bargaining with gendered threat points, doing gender and patrilocal norms all predict this.

H2 (sources of the additional unpaid time). Specialisation predicts that women's additional

unpaid work is financed mainly by lower paid work and learning, so total work changes little (H2a). A time-squeeze or greedy-institution account predicts that it is financed mainly by lower leisure, social time and self-care, including sleep, so total work rises (H2b). Because sleep responds to the value of time but has a biological floor (Basner et al., 2007; Biddle & Hamermesh, 1990), I expect sleep to adjust less than free time.

H3 (patrilocal co-residence). If senior women share domestic work, daughters-in-law of the head do less unpaid work than wives in households without a parent generation (H3a, substitution). If the household hierarchy assigns the heaviest work to the junior wife and restricts her movement (Anukriti et al., 2020; Kandiyoti, 1988), daughters-in-law do more unpaid work, less paid work and less social time outside the home (H3b, hierarchy).

H4 (relative resources within couples). Bargaining and relative-resources theories predict that the wife–husband gap in unpaid work is smaller when the wife is employed, when she is more educated than her husband, and when the spousal age gap is smaller. Age gaps proxy for seniority in a hierarchical marriage and are larger when women marry young (Desai & Andrist, 2010). Doing-gender accounts predict weaker, or non-monotonic, associations with the wife's relative position (Brines, 1994; Greenstein, 2000).

Method

Data

I use the 2024 round because it is the most recent and its diaries cover every household member aged six and above; the 2019 round (National Statistical Office, 2020) uses the

same design, and with a harmonised specification (age, state, diary day and residence) the married–never-married difference in women's unpaid work is 284.4 minutes (SE 2.6) in 2019 and 291.7 minutes (SE 2.5) in 2024, and for men 15.7 minutes (SE 1.0) and 25.7 minutes (SE 1.0), so the pattern is not specific to one round. The Time Use Survey 2024, conducted by India's National Statistics Office from January to December 2024, records the activities of every household member aged six and above over the 24 hours from 4:00 a.m. on the day before the interview (National Statistics Office, 2025). Activities are coded to the three-digit level of ICATUS 2016 (United Nations Statistics Division, 2016). Where activities overlap, the survey flags one as the major activity. I use every activity record (10,211,478 records from 450,457 diaries in 139,487 households) and aggregate major activities to the person-day, so all but 5 of the 450,457 diaries partition exactly 1,440 minutes. The roster lists 454,192 persons aged six and above; 3,735 of them have no diary records in the file, which leaves the 450,457 diaries, and the 49 diaries of third-gender respondents are excluded from the comparisons of women and men. Some diaries are reported by another household member; I control for proxy reporting and drop proxy diaries in a robustness check. All estimates use the survey multipliers, and standard errors are clustered on the first-stage unit, the primary sampling unit (Abadie et al., 2023; Solon et al., 2015).

Samples

The main sample consists of never-married and currently married women and men aged 20 to 39 (88,337 women and 87,364 men). Widowed, divorced and separated persons are excluded from the main comparison and studied separately at ages 50 to 79. In the main age range,

85% of women and 65% of men are currently married.

The couple sample uses the household roster to link spouses. A head is linked to the spouse of the head, and a married child of the head is linked to the child's spouse when each is unique in the household and the two are of opposite sex. 89% of married persons in the survey have a linked spouse diary. Couples with several married children of the head in the same household are not linked, which under-represents the largest joint households. I keep linked wives aged 20 to 59 (106,858 couples; 106,424 have complete covariates and enter the couple regressions). Household members' diaries are collected in the same visit; in 100% of linked couples the two diaries refer to the same day of the week.

Measures

The outcomes are minutes per day in the nine ICATUS divisions: employment and related activities (including work travel), production of goods for own final use (for example farming for the household, fetching water and collecting firewood), unpaid domestic services, unpaid caregiving for household members, volunteering, learning, socialising and religious practice, culture and leisure (including mass media and sport), and self-care and maintenance (including sleep, eating and personal care). Unpaid work is the sum of domestic services and caregiving, the time-use measure behind Sustainable Development Goal indicator 5.4.1. Sub-rows report sleep, socialising (talking and getting together), exercise and sport, religious practice and mass media.

Marital status and relationship to the household head come from the roster. For married women, I distinguish three residence types: (a) the woman is the head or the head's wife and

no parent or parent-in-law of the head lives in the household ("no parent generation"), (b) she is the head or the head's wife and a parent or parent-in-law of the head is present, and (c) she is the daughter-in-law of the head (roster code 4). Among married women aged 20 to 39, these groups account for 60%, 8% and 29% respectively; the remaining 3% hold other positions (for example, a married daughter of the head). For men, the junior category is a married son of the head (code 3). Type (c) is the clearest indicator of patrilocal co-residence. Type (b) mixes households in which the parent generation is the husband's and households in which it is the wife's, because the roster records the relation to the head only.

Covariates are single-year age, education (four tiers), urban residence, caste group (Scheduled Tribe, Scheduled Caste, Other Backward Class, other), religion, state, day of the week and proxy reporting. The baseline model does not control for household size, older household members or consumption. These are partly consequences of marriage, since at marriage a woman usually moves into a different and often larger household, and controlling for them would remove part of the association I want to measure. They enter an extended specification as a robustness check.

Analytic strategy

For each activity division k, separately for women and men,

$$y_i^k = \beta^k \,\mathrm{Married}_i + X_i'\gamma^k + \alpha_{a(i)} + \mu_{s(i)} + \tau_{d(i)} + \varepsilon_i^k,$$

where α, μ and τ are single-year age, state and day-of-week fixed effects. Because the nine

divisions sum to 1,440 minutes for all but 5 of the 450,457 diaries and every regression uses the same sample, weights and regressors, $\sum_k \widehat{\beta}^k = 0$ up to rounding. The coefficients decompose the adjusted married–never-married difference into its sources. A pooled model with a married × woman interaction and sex-specific age, state and day effects tests whether the unpaid-work difference is gendered. Heterogeneity by residence replaces the married indicator with indicators for the three residence types, each compared with the same never-married group. A model restricted to married women tests whether daughters-in-law differ from wives in households without a parent generation.

The coefficient $\widehat{\beta}^k$ is not the causal effect of marriage. The survey is a single cross-section, it does not record the date of marriage, and the never-married are a selected and shrinking group. I use four strategies to judge how much selection could matter. (1) I re-estimate the decomposition for women aged 20 to 26, where 32% of women are never married and the comparison group is closest to the population. (2) I reweight never-married respondents to the joint distribution of the covariates among the married (DiNardo et al., 1996). The weights are the odds of being married from a logit on single-year age, state, education, urban residence, caste and religion (Imbens, 2004; Rosenbaum & Rubin, 1983), with propensities capped at 0.99. The reweighted regression, which also includes the controls, estimates the average difference among the married, conditional on observables, and is doubly robust to misspecification of either the propensity or the outcome model. (3) For women, I compute bias-adjusted coefficients that assume selection on unobservables is as strong as selection on observables ($\delta = 1$) and that the maximum attainable R^2 is 1.3 times the controlled R^2, and report the value of δ at which the adjusted coefficient would reach zero (Altonji et al., 2005;

Oster, 2019). (4) I estimate the unpaid-work gap separately in five age bands. If selection of the never-married drives the gap, the gap should grow sharply as the never-married become rarer.

Within-couple comparisons difference out everything shared by the spouses: household composition, consumption, location, the diary day and much of the social environment. I report the wife-minus-husband gap in unpaid work, the wife's share of the couple's unpaid work, and the husband-minus-wife gap in employment time, by the wife's employment, relative education, residence type and the spousal age gap. Adjusted associations for the age gap and residence type control for the wife's single-year age, both spouses' education, caste, religion, urban residence, state and day of the week. These comparisons describe the division of labour within marriage, not the effect of marriage, and the wife's employment is itself a joint decision of the couple and household.

Results

Descriptive patterns

Table 1 describes the main sample. Never-married women in this age range are much more likely to be students (32%, against 0% of married women), and 28% of never-married women report being employed in their usual principal status, against 22% of married women. The raw differences between columns mix the association with marriage and the different age and education profiles of the groups, which the regressions below hold constant.

Table 1: Weighted means by sex and marital status, ages 20–39

	Women, never married	Women, married	Men, never married	Men, married
Employment (incl. travel), min	120	70	323	452
Own-use production of goods, min	12	23	10	19
Unpaid domestic services, min	164	356	29	26
Unpaid caregiving, min	5	103	2	35
Learning, min	154	2	92	2
Socialising, community, religion, min	91	90	96	87
Culture, leisure, media, sport, min	169	112	179	118
Self-care incl. sleep, min	723	684	709	699
of which: sleep, min	544	527	527	515
Age, years	23.4	29.8	24.8	31.7
Employed (usual status)	28%	22%	68%	98%
Current student	32%	0%	19%	0%
Graduate	45%	16%	32%	20%
Urban	43%	29%	41%	31%
Diaries (N)	13,940	74,397	31,598	55,766

Notes: Minutes per day from major activities; the nine divisions (eight shown plus volunteering) sum to 1,440. Weighted by survey multipliers.

Decomposing the married–never-married difference

Table 2: Married minus never-married, minutes per day, ages 20–39

Activity	Women	Men
Employment (incl. travel)	-84.7*** (3.3)	69.4*** (2.7)
Own-use production of goods	-1.0 (0.7)	2.4*** (0.6)
Unpaid domestic services	164.8*** (2.3)	-5.5*** (0.8)
Unpaid caregiving	114.9*** (1.1)	31.7*** (0.5)
Volunteering	-0.2 (0.2)	-0.8*** (0.2)
Learning	-118.7*** (2.6)	-36.1*** (1.2)
Socialising, community, religion	-6.6*** (1.4)	-11.0*** (1.2)
Culture, leisure, media, sport	-41.3*** (1.7)	-42.2*** (1.3)
Self-care incl. sleep	-27.1*** (1.6)	-7.9*** (1.3)
Sum of the nine divisions	0.0	-0.0
of which: sleep	-8.6*** (1.4)	-8.7*** (1.1)
of which: socialising	-8.8*** (1.1)	-11.8*** (1.0)
of which: exercise and sport	-2.2*** (0.2)	-3.5*** (0.3)
of which: religious practice	2.7*** (0.5)	0.5 (0.4)
of which: mass media	-35.1*** (1.4)	-25.8*** (1.0)
Unpaid work (domestic + care)	279.7*** (2.5)	26.2*** (0.9)

Notes: Coefficients on currently married from separate regressions by sex, with single-year age, state and day-of-week fixed effects and controls for education, urban residence, caste, religion and proxy report. FSU-clustered standard errors in parentheses. The nine divisions partition the day, so their coefficients sum to zero. * p<0.10, ** p<0.05, *** p<0.01.

Table 2 and Figure S1 report the decomposition. Marriage is associated with 164.8 minutes (SE 2.3) more unpaid domestic work and 114.9 minutes (SE 1.1) more caregiving for women.

For men the corresponding coefficients are -5.5 minutes (SE 0.8) and 31.7 minutes (SE 0.5). In the pooled model, the married × woman coefficient on unpaid work is 261.9 minutes (SE 2.6) (statistically significant at the 1% level). The pattern supports H1: marriage is associated with much more unpaid work for women than for men.

Part of this difference reflects motherhood rather than marriage as such: married women are far more likely to live with young children, and caregiving accounts for 115 of the 280 minutes. Among women in households with no child under six (38,541 women, 70% of them married), the married–never-married difference in unpaid work is 186.6 minutes (SE 3.0): 175.5 minutes (SE 2.9) in domestic work and 11.1 minutes (SE 0.5) in caregiving. The domestic-work difference largely persists without young children in the household, whereas most of the caregiving difference does not. The estimates are therefore best read as the difference that accompanies marriage, including the motherhood that usually follows it, not as what marriage alone adds.

Among the other seven divisions, women's additional unpaid minutes are offset mainly by lower time in learning (119 minutes), employment (incl. travel) (85 minutes) and culture, leisure, media, sport (41 minutes). Employment and learning together account for 73% of the reductions, socialising and leisure for 17%, and self-care for 10%. Most of the offset comes from paid work and learning, which is closer to the specialisation account (H2a) than to a squeeze on free time.

The sub-rows place the commonly studied "marriage penalties" within the budget. Married women slept 9 minutes less than never-married women (SE 1.4). The coefficients for socialising, exercise and sport, and religious practice are -8.8 minutes (SE 1.1), -2.2 minutes (SE

0.2) and 2.7 minutes (SE 0.5). Sleep moves less than free time, as H2 anticipated. For men, the largest coefficient in absolute value is on employment (incl. travel), at 69.4 minutes (SE 2.7).

Patrilocal co-residence

Table S1 splits married women and men by residence type. Each married group is compared with never-married persons of the same sex, so each column again sums to zero across divisions. The comparison of interest is between daughters-in-law of the head and wives in households without a parent generation. In a model restricted to married women, daughters-in-law do 2.1 minutes (SE 1.8) less unpaid work (not statistically distinguishable from zero), 12.6 minutes (SE 1.7) less employment, 0.8 minutes (SE 1.1) more socialising, community and religious activity, and sleep 1.1 minutes (SE 1.1) more. Unpaid work does not differ detectably by residence type, so the data support neither the substitution account (H3a) nor the hierarchy account (H3b) as a description of average unpaid time.

Two cautions apply. Daughters-in-law are younger on average and at an earlier stage of marriage than wives who head their own household. The single-year age effects absorb the age difference, but not marital duration, which the survey does not record. In addition, the move from a joint to a nuclear household is itself selective, and can follow the death of the senior couple or household partition (Debnath, 2015).

Table 3: Women: married minus never-married under alternative treatments of selection, minutes per day

Activity	Baseline	Ages 20–26	Reweighted	Bias-adjusted	Breakdown delta
Employment (incl. travel)	-84.7*** (3.3)	-68.3*** (3.2)	-72.0*** (6.1)	-96.2	-7.40
Own-use production of goods	-1.0 (0.7)	0.3 (0.8)	3.4* (1.8)		
Unpaid domestic services	164.8*** (2.3)	161.6*** (2.6)	141.6*** (5.3)		
Unpaid caregiving	114.9*** (1.1)	105.4*** (1.4)	91.4*** (1.7)		
Volunteering	-0.2 (0.2)	-0.0 (0.3)	-0.1 (0.3)		
Learning	-118.7*** (2.6)	-131.2*** (3.0)	-30.7*** (1.5)	-43.0	1.57
Socialising, community, religion	-6.6*** (1.4)	-4.4*** (1.7)	-18.7*** (2.9)	-8.2	-4.30
Culture, leisure, media, sport	-41.3*** (1.7)	-38.0*** (2.0)	-67.9*** (3.8)	-33.1	5.04
Self-care incl. sleep	-27.1*** (1.6)	-25.3*** (1.9)	-47.1*** (3.4)		
of which: sleep	-8.6*** (1.4)	-8.9*** (1.6)	-26.9*** (2.8)	-5.7	3.01
Unpaid work (domestic + care)	279.7*** (2.5)	267.0*** (2.8)	233.0*** (5.5)	264.3	18.14

Notes: Baseline: the decomposition for ages 20–39. Ages 20–26: same model on women aged 20–26. Reweighted: never-married women reweighted to the covariate distribution of married women (DiNardo-Fortin-Lemieux odds weights from a logit on single-year age, state, education, urban, caste and religion; propensities capped at 0.99), with the baseline controls. Bias-adjusted: Oster (2019) coefficient with delta = 1 and R-max = 1.3 times the controlled R-squared, relative to a regression on the married indicator alone. Breakdown delta: the delta at which the bias-adjusted coefficient is zero. FSU-clustered SEs in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Selection into marriage

Reweighting leans on a small number of never-married women who resemble the married: the effective sample of never-married women falls to 1,209 from 13,940. Trimming observations with propensities outside [0.05, 0.95] gives an unpaid-work difference of 270.5 minutes (SE 3.1) (effective never-married sample 3,006), and outside [0.10, 0.90] 268.6 minutes (SE 3.1) (3,842) (Crump et al., 2009). The learning and employment coefficients move more across these weightings (learning -70.5 and -88.7; employment -106.9 and -98.1), so the reweighted estimates for those divisions are imprecise guides. Figure S2 in the online supplement shows the overlap of propensity scores.

Table 3 reports four checks for women. Among women aged 20 to 26, the unpaid-work coefficient is 267.0 minutes (SE 2.8), against 279.7 minutes (SE 2.5) in the baseline. After reweighting, the never-married resemble the married on observed characteristics: the largest absolute standardised difference across the covariates in Table S2 in the online supplement

is 0.13, and Table S3 in the online supplement reports the coefficient-stability calculation in full. The reweighted, regression-adjusted coefficient is 233.0 minutes (SE 5.5) (without controls, 248.7 minutes (SE 5.8)). Reweighting concentrates weight on the never-married women who most resemble the married, so the effective sample of never-married women falls from 13,940 to about 1,209, and 2.0% of the weighted never-married have propensities above the cap. The bias-adjusted coefficient on unpaid work is 264.3 minutes. Selection on unobservables would have to be 18.1 times as strong as selection on observables to explain away the unpaid-work difference.

Figure S3 shows the unpaid-work gap by age band. Among women, the coefficient is 256.10 minutes (SE 3.97) at ages 20 to 22, where 53% are never married, and 214.7 minutes (SE 10.8) at ages 35 to 39, where 2% are. The gap does not grow significantly as the never-married become rarer, which is not what strong positive selection of the never-married on unpaid-work preferences would produce. Comparisons at older ages also mix the association with marriage and the accumulation of children, which the survey links to mothers only partly.

Within married couples

Within couples, the household and the diary day are held constant. Table 4 shows that wives do 366.9 minutes (SE 1.1) more unpaid work than their husbands and account for 91% of the couple's unpaid minutes; husbands do 336.4 minutes (SE 1.5) more employment, and have 12.9 minutes (SE 0.9) more free time than their wives.

The wife–husband gap in unpaid work is 258 minutes when the wife is employed and 408

Table 4: Within-couple gaps, linked couples with wives aged 20–59

	Unpaid gap (W−H)	Wife's unpaid share	Employment gap (H−W)	N
All linked couples	366.9 (1.1)	91% (0.1)	336.4 (1.5)	106,424
Wife's employment				
Wife not employed	408.1 (1.1)	91% (0.1)	411.4 (1.5)	77,619
Wife employed	257.9 (1.4)	90% (0.1)	138.3 (2.0)	28,805
Relative education				
Husband more educated	367.3 (1.4)	91% (0.1)	337.1 (2.1)	32,345
Same education tier	362.7 (1.2)	91% (0.1)	331.4 (1.7)	59,896
Wife more educated	383.0 (1.8)	91% (0.1)	355.1 (2.9)	14,183
Residence				
Head couple, no parent generation	356.5 (1.1)	91% (0.1)	328.1 (1.6)	80,850
Head couple, parent generation present	376.3 (2.3)	92% (0.2)	328.1 (3.3)	8,177
Wife is daughter-in-law of head	414.4 (1.9)	92% (0.1)	382.3 (2.4)	16,965
Spousal age gap				
Husband same age or younger	369.7 (3.9)	91% (0.3)	351.2 (6.4)	3,189
Husband older by 1-3 years	375.8 (1.5)	91% (0.1)	348.3 (2.0)	42,460
Husband older by 4-6 years	364.3 (1.4)	91% (0.1)	339.2 (2.0)	38,268
Husband older by 7-9 years	359.0 (1.9)	91% (0.1)	318.7 (2.8)	13,651
Husband older by 10+ years	345.6 (2.2)	90% (0.2)	288.1 (3.6)	8,856

Notes: Spouses linked through the household roster (head and spouse of head; married child of head and child's spouse where each is unique). Weighted by the wife's multiplier. FSU-clustered SEs in parentheses (share SE in percentage points).

minutes when she is not, a difference of -150.2 minutes (SE 1.6). The gap in couples where the wife has more education than her husband differs from the gap in couples where the husband has more education by 15.7 minutes (SE 2.1). The gap is smaller when the wife is employed but not when she is more educated, so the evidence for H4 is mixed. Even in couples where the wife is employed, the gap remains more than half its average size. These comparisons are descriptive. Employment is chosen jointly with the division of housework, and employed wives also differ in age, education and household type.

In households where the wife is the daughter-in-law of the head, the wife–husband gap differs from that in households without a parent generation by 58.0 minutes (SE 1.9) unadjusted and 11.8 minutes (SE 1.9) after adjustment. The median husband in linked couples is 4 years older than his wife. Figure S4 plots the unpaid-work gap by the spousal age gap. Net of the wife's age, both spouses' education and location, each additional year by which the husband is older is associated with a gap that is 0.92 minutes (SE 0.22) larger, and with a

wife's share that is 0.04 percentage points (SE 0.02) lower per year. Unadjusted, the gap is 370 minutes where the husband is the same age or younger and 346 where he is ten or more years older (Table 4), so in raw terms the division is less unequal when the husband is much older; large age gaps are concentrated among older wives, whose unpaid work is lower. Net of the wife's age, the minute gap is larger but the wife's share is smaller, so the evidence on the seniority interpretation of H4 is mixed. Because larger age gaps are more common when women marry young, the age gap also stands in for the wife's age at marriage and its correlates (Desai & Andrist, 2010; Field & Ambrus, 2008).

Widowhood at older ages

Table S4 applies the same accounting to the end of marriage. Widowed women do 73.0 minutes (SE 1.7) less unpaid work than married women of the same age, and widowed men 43.9 minutes (SE 1.7) more than married men. This comparison is also selective, since widowhood is related to the spouse's health, to poverty and to the age gap between spouses, and many widowed Indians live with adult children. It shows which activities differ when a spouse is no longer present, not what widowhood causes.

Robustness

Table S5 re-estimates the unpaid-work coefficient for women and men under alternative samples and controls. Across specifications, the coefficient for women ranges from 256.1 to 297.2 minutes. The lowest, for ages 30 to 39 (256.14, SE 6.24), coincides at one decimal with the age 20 to 22 band coefficient (256.10, SE 3.97); the two are separate estimates on

different samples. It is positive and statistically significant in every specification. Excluding current students removes the group whose time is most mechanically devoted to learning, but student status is itself affected by marriage, so this row is a bound rather than a preferred estimate. Adding household size, older members and consumption changes the coefficient for women from 279.7 to 277.0 minutes, consistent with part of the association running through the household a woman joins at marriage.

Discussion

Because the day is fixed, more unpaid work must coincide with less of something else. This paper replaces separate estimates of marriage "penalties" in housework, sleep, social life and exercise with one decomposition in which the parts must add up. In India in 2024, married women aged 20 to 39 do substantially more unpaid domestic and care work than never-married women, and the difference is far larger for women than for men. The offsetting differences are mostly in paid work and learning, which fits the specialisation account, but specialisation in India happens against a background of low and falling married women's employment and strong norms about women's work outside the home, so it is not necessarily a choice made within the couple.

The findings speak to three debates. First, single-outcome studies can find a sleep or leisure penalty without showing whether it is the largest adjustment; the decomposition makes the trade-offs explicit for women and men. Second, on patrilocality, daughters-in-law do not differ detectably from other wives in unpaid minutes (Anukriti et al., 2020; Khalil & Mookerjee, 2019). Third, on resources within couples, the gap is smaller where the wife is

employed, as bargaining theories predict, but it remains large in every group in Table 4, consistent with gender operating through institutions and interaction as well as individual resources (Ridgeway & Correll, 2004; Risman, 2004).

Limitations

The main limitation is the comparison group. Married and never-married women differ in ways that age, education and location do not capture, and the survey does not record when a marriage began. The four selection checks narrow the range of plausible interpretations but do not identify the causal effect of marriage. The estimates are best read as the time-use profile that accompanies marriage in contemporary India, with the restricted-age and reweighted estimates as the most credible magnitudes. Second, the couple analysis covers only spouses whom the roster can link without ambiguity. Joint households with several married sons are excluded, and these are probably the households in which hierarchy among daughters-in-law matters most. Third, the residence categories describe relationships to the household head, not to the woman's own parents-in-law, and the "parent generation present" category mixes the husband's and the wife's parents. Fourth, a single diary day per person measures daily time use with error. This adds noise but should not bias the comparisons of means, except where the diary day differs systematically by marital status; the day-of-week effects and the normal-day check address the most obvious source. Fifth, parenthood is a principal channel through which marriage changes time use (Baxter et al., 2008; Sanchez & Thomson, 1997). The check on women without a child under six separates part of it, but older children remain; panel data on the timing of marriage and births would allow an

event-study design in the style of Kleven et al. (2019).

Conclusion

Married and never-married Indian women spend the day very differently. Married women do far more unpaid domestic and care work, a difference with no counterpart among men. A time-budget decomposition shows which activities differ alongside that work, and the household roster shows how the change depends on whether a young wife joins her husband's parents' household and on her standing relative to her husband. Policies that aim to raise married women's employment, such as childcare, safe transport or flexible work, operate within this budget. They are more likely to succeed if they also reduce the unpaid domestic and care work that accompanies marriage and motherhood, since that work competes with women's paid work, learning, leisure or sleep.

Declaration of generative AI use

Generative AI tools (Claude, Anthropic) were used to assist with writing and debugging analysis code, checking references, and drafting and editing text. The author reviewed and verified all analyses, results, and text, and takes full responsibility for the content.

Data and code availability

The Time Use Survey 2024 unit-level data are distributed by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India (microdata.gov.in, catalogue 236), to registered users. Code available from the author on request; will be made public on acceptance.

References

Abadie, A., Athey, S., Imbens, G. W., & Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *Quarterly Journal of Economics*, *138*(1), 1–35.

Afridi, F., Dinkelman, T., & Mahajan, K. (2018). Why are fewer married women joining the work force in rural India? A decomposition analysis over two decades. *Journal of Population Economics*, *31*(3), 783–818. https://doi.org/10.1007/s00148-017-0671-y

Agarwal, B. (1997). "Bargaining" and gender relations: Within and beyond the household. *Feminist Economics*, *3*(1), 1–51. https://doi.org/10.1080/135457097338799

Aguiar, M., & Hurst, E. (2007). Measuring trends in leisure: The allocation of time over five decades. *Quarterly Journal of Economics*, *122*(3), 969–1006. https://doi.org/10.116

2/qjec.122.3.969

Akerlof, G. A., & Kranton, R. E. (2000). Economics and identity. *Quarterly Journal of Economics*, *115*(3), 715–753. https://doi.org/10.1162/003355300554881

Allendorf, K., & Pandian, R. K. (2016). The decline of arranged marriage? Marital change and continuity in India. *Population and Development Review*, *42*(3), 435–464. https://doi.org/10.1111/j.1728-4457.2016.00149.x

Altonji, J. G., Elder, T. E., & Taber, C. R. (2005). Selection on observed and unobserved variables: Assessing the effectiveness of Catholic schools. *Journal of Political Economy*, *113*(1), 151–184. https://doi.org/10.1086/426036

Anukriti, S., Herrera-Almanza, C., Pathak, P. K., & Karra, M. (2020). Curse of the mummy-ji: The influence of mothers-in-law on women in India. *American Journal of Agricultural Economics*, *102*(5), 1328–1351.

Arber, S., Bote, M., & Meadows, R. (2009). Gender and socio-economic patterning of self-reported sleep problems in Britain. *Social Science & Medicine*, *68*(2), 281–289. https://doi.org/10.1016/j.socscimed.2008.10.016

Basner, M., Fomberstein, K. M., Razavi, F. M., Banks, S., William, J. H., Rosa, R. R., & Dinges, D. F. (2007). American Time Use Survey: Sleep time and its relationship to waking activities. *Sleep*, *30*(9), 1085–1095. https://doi.org/10.1093/sleep/30.9.1085

Baxter, J., Hewitt, B., & Haynes, M. (2008). Life course transitions and housework: Marriage, parenthood, and time on housework. *Journal of Marriage and Family*, *70*(2), 259–272. https://doi.org/10.1111/j.1741-3737.2008.00479.x

Becker, G. S. (1965). A theory of the allocation of time. *Economic Journal*, *75*(299), 493–517. https://doi.org/10.2307/2228949

Becker, G. S. (1985). Human capital, effort, and the sexual division of labor. *Journal of Labor Economics*, *3*(1, Part 2), S33–S58. https://doi.org/10.1086/298075

Becker, G. S. (1991). *A treatise on the family* (Enlarged). Harvard University Press. https://doi.org/10.4159/9780674020665

Bertrand, M., Kamenica, E., & Pan, J. (2015). Gender identity and relative income within households. *Quarterly Journal of Economics*, *130*(2), 571–614. https://doi.org/10.1093/qje/qjv001

Bianchi, S. M., Milkie, M. A., Sayer, L. C., & Robinson, J. P. (2000). Is anyone doing the housework? Trends in the gender division of household labor. *Social Forces*, *79*(1), 191–228. https://doi.org/10.2307/2675569

Biddle, J. E., & Hamermesh, D. S. (1990). Sleep and the allocation of time. *Journal of Political Economy*, *98*(5), 922–943. https://doi.org/10.1086/261713

Bittman, M., England, P., Sayer, L., Folbre, N., & Matheson, G. (2003). When does gender trump money? Bargaining and time in household work. *American Journal of Sociology*, *109*(1), 186–214. https://doi.org/10.1086/378341

Bittman, M., & Wajcman, J. (2000). The rush hour: The character of leisure time and gender equity. *Social Forces*, *79*(1), 165–189. https://doi.org/10.2307/2675568

Blood, R. O., & Wolfe, D. M. (1960). *Husbands and wives: The dynamics of married living*. Free Press.

Brines, J. (1994). Economic dependency, gender, and the division of labor at home. *American Journal of Sociology*, *100*(3), 652–688. https://doi.org/10.1086/230577

Burda, M., Hamermesh, D. S., & Weil, P. (2013). Total work and gender: Facts and possible explanations. *Journal of Population Economics*, *26*(1), 239–261. https://doi.org/10.100

7/s00148-012-0408-x

Burgard, S. A., & Ailshire, J. A. (2013). Gender and time for sleep among U.S. adults. *American Sociological Review*, *78*(1), 51–69. https://doi.org/10.1177/0003122412472048

Chatterjee, E., Desai, S., & Vanneman, R. (2018). Indian paradox: Rising education, declining women's employment. *Demographic Research*, *38*, 855–878. https://doi.org/10.4054/demres.2018.38.31

Chiappori, P.-A. (1992). Collective labor supply and welfare. *Journal of Political Economy*, *100*(3), 437–467. https://doi.org/10.1086/261825

Coltrane, S. (2000). Research on household labor: Modeling and measuring the social embeddedness of routine family work. *Journal of Marriage and Family*, *62*(4), 1208–1233. https://doi.org/10.1111/j.1741-3737.2000.01208.x

Coser, L. A. (1974). *Greedy institutions: Patterns of undivided commitment*. Free Press.

Crump, R. K., Hotz, V. J., Imbens, G. W., & Mitnik, O. A. (2009). Dealing with limited overlap in estimation of average treatment effects. *Biometrika*, *96*(1), 187–199. https://doi.org/10.1093/biomet/asn055

Davis, S. N., & Greenstein, T. N. (2009). Gender ideology: Components, predictors, and consequences. *Annual Review of Sociology*, *35*, 87–105. https://doi.org/10.1146/annurev-soc-070308-115920

Debnath, S. (2015). The impact of household structure on female autonomy in developing countries. *Journal of Development Studies*, *51*(5), 485–502. https://doi.org/10.1080/00220388.2014.983909

Desai, S., & Andrist, L. (2010). Gender scripts and age at marriage in India. *Demography*, *47*(3), 667–687. https://doi.org/10.1353/dem.0.0118

Dhanaraj, S., & Mahambare, V. (2019). Family structure, education and women's employment in rural India. *World Development*, *115*, 17–29. https://doi.org/10.1016/j.worlddev.2018.11.004

DiNardo, J., Fortin, N. M., & Lemieux, T. (1996). Labor market institutions and the distribution of wages, 1973–1992: A semiparametric approach. *Econometrica*, *64*(5), 1001–1044. https://doi.org/10.2307/2171954

Dyson, T., & Moore, M. (1983). On kinship structure, female autonomy, and demographic behavior in India. *Population and Development Review*, *9*(1), 35–60. https://doi.org/10.2307/1972894

Eswaran, M., Ramaswami, B., & Wadhwa, W. (2013). Status, caste, and the time allocation of women in rural India. *Economic Development and Cultural Change*, *61*(2), 311–333. https://doi.org/10.1086/668282

Field, E., & Ambrus, A. (2008). Early marriage, age of menarche, and female schooling attainment in Bangladesh. *Journal of Political Economy*, *116*(5), 881–930. https://doi.org/10.1086/593333

Field, E., Pande, R., Rigol, N., Schaner, S., & Moore, C. T. (2021). On her own account: How strengthening women's financial control impacts labor supply and gender norms. *American Economic Review*, *111*(7), 2342–2375. https://doi.org/10.1257/aer.20200705

Fletcher, E., Pande, R., & Moore, C. T. (2017). *Women and work in India: Descriptive evidence and a review of potential policies* (Faculty Research Working Paper Nos. RWP18-004). Harvard Kennedy School.

Fuwa, M. (2004). Macro-level gender inequality and the division of household labor in 22 countries. *American Sociological Review*, *69*(6), 751–767. https://doi.org/10.1177/0003

12240406900601

Gershuny, J. (2000). *Changing times: Work and leisure in postindustrial society*. Oxford University Press.

Gimenez-Nadal, J. I., & Sevilla, A. (2012). Trends in time allocation: A cross-country analysis. *European Economic Review*, *56*(6), 1338–1359. https://doi.org/10.1016/j.euroecorev.2012.02.011

Goldman, N. (1993). Marriage selection and mortality patterns: Inferences and fallacies. *Demography*, *30*(2), 189–208. https://doi.org/10.2307/2061837

Greenstein, T. N. (2000). Economic dependence, gender, and the division of labor in the home: A replication and extension. *Journal of Marriage and Family*, *62*(2), 322–335. https://doi.org/10.1111/j.1741-3737.2000.00322.x

Gronau, R. (1977). Leisure, home production, and work—the theory of the allocation of time revisited. *Journal of Political Economy*, *85*(6), 1099–1123. https://doi.org/10.1086/260629

Gupta, S. (1999). The effects of transitions in marital status on men's performance of housework. *Journal of Marriage and Family*, *61*(3), 700–711. https://doi.org/10.2307/353571

Gupta, S. (2007). Autonomy, dependence, or display? The relationship between married women's earnings and housework. *Journal of Marriage and Family*, *69*(2), 399–417. https://doi.org/10.1111/j.1741-3737.2007.00373.x

Hirway, I., & Jose, S. (2011). Understanding women's work using time-use statistics: The case of India. *Feminist Economics*, *17*(4), 67–92. https://doi.org/10.1080/13545701.2011.622289

Hook, J. L. (2006). Care in context: Men's unpaid work in 20 countries, 1965–2003. *American Sociological Review*, *71*(4), 639–660. https://doi.org/10.1177/000312240607100406

Hook, J. L. (2010). Gender inequality in the welfare state: Sex segregation in housework, 1965–2003. *American Journal of Sociology*, *115*(5), 1480–1523. https://doi.org/10.1086/651384

Imbens, G. W. (2004). Nonparametric estimation of average treatment effects under exogeneity: A review. *Review of Economics and Statistics*, *86*(1), 4–29. https://doi.org/10.1162/003465304323023651

Janiso, A., Shukla, P. K., & Bheemeshwar Reddy A. (2024). What explains the gender gap in unpaid housework and care work in India? *Development Policy Review*, *42*(1), e12730. https://doi.org/10.1111/dpr.12730

Jayachandran, S. (2021). Social norms as a barrier to women's employment in developing countries. *IMF Economic Review*, *69*(3), 576–595. https://doi.org/10.1057/s41308-021-00140-w

Jejeebhoy, S. J., & Sathar, Z. A. (2001). Women's autonomy in India and Pakistan: The influence of religion and region. *Population and Development Review*, *27*(4), 687–712. https://doi.org/10.1111/j.1728-4457.2001.00687.x

Jensen, R. (2012). Do labor market opportunities affect young women's work and family decisions? Experimental evidence from India. *Quarterly Journal of Economics*, *127*(2), 753–792. https://doi.org/10.1093/qje/qjs002

Juster, F. T., & Stafford, F. P. (1991). The allocation of time: Empirical findings, behavioral models, and problems of measurement. *Journal of Economic Literature*, *29*(2), 471–522.

Kan, M. Y., Sullivan, O., & Gershuny, J. (2011). Gender convergence in domestic work:

Discerning the effects of interactional and institutional barriers from large-scale data. *Sociology*, *45*(2), 234–251. https://doi.org/10.1177/0038038510394014

Kandiyoti, D. (1988). Bargaining with patriarchy. *Gender & Society*, *2*(3), 274–290. https://doi.org/10.1177/089124388002003004

Khalil, U., & Mookerjee, S. (2019). Patrilocal residence and women's social status: Evidence from South Asia. *Economic Development and Cultural Change*, *67*(2), 401–438. https://doi.org/10.1086/697584

Killewald, A., & Gough, M. (2010). Money isn't everything: Wives' earnings and housework time. *Social Science Research*, *39*(6), 987–1003. https://doi.org/10.1016/j.ssresearch.2010.08.005

Klasen, S., & Pieters, J. (2015). What explains the stagnation of female labor force participation in urban India? *World Bank Economic Review*, *29*(3), 449–478.

Kleven, H., Landais, C., & Søgaard, J. E. (2019). Children and gender inequality: Evidence from Denmark. *American Economic Journal: Applied Economics*, *11*(4), 181–209. https://doi.org/10.1257/app.20180010

Lachance-Grzela, M., & Bouchard, G. (2010). Why do women do the lion's share of housework? A decade of research. *Sex Roles*, *63*(11), 767–780. https://doi.org/10.1007/s11199-010-9797-z

Lundberg, S., & Pollak, R. A. (1993). Separate spheres bargaining and the marriage market. *Journal of Political Economy*, *101*(6), 988–1010. https://doi.org/10.1086/261912

Manser, M., & Brown, M. (1980). Marriage and household decision-making: A bargaining analysis. *International Economic Review*, *21*(1), 31–44. https://doi.org/10.2307/2526238

Mattingly, M. J., & Bianchi, S. M. (2003). Gender differences in the quantity and quality of free time: The U.S. experience. *Social Forces*, *81*(3), 999–1030. https://doi.org/10.1353/sof.2003.0036

Maume, D. J., Sebastian, R. A., & Bardo, A. R. (2010). Gender, work-family responsibilities, and sleep. *Gender & Society*, *24*(6), 746–768. https://doi.org/10.1177/0891243210386949

McElroy, M. B., & Horney, M. J. (1981). Nash-bargained household decisions: Toward a generalization of the theory of demand. *International Economic Review*, *22*(2), 333–349. https://doi.org/10.2307/2526280

Mehrotra, S., & Parida, J. K. (2017). Why is the labour force participation of women declining in India? *World Development*, *98*, 360–380. https://doi.org/10.1016/j.worlddev.2017.05.003

National Statistical Office. (2020). *Time use in india 2019*. Ministry of Statistics; Programme Implementation, Government of India.

National Statistics Office. (2025). *Time use survey (January–December 2024)*. Ministry of Statistics; Programme Implementation, Government of India.

Nomaguchi, K. M., & Bianchi, S. M. (2004). Exercise time: Gender differences in the effects of marriage, parenthood, and employment. *Journal of Marriage and Family*, *66*(2), 413–430. https://doi.org/10.1111/j.1741-3737.2004.00029.x

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, *37*(2), 187–204. https://doi.org/10.1080/07350015.2016.1227711

Rahman, L., & Rao, V. (2004). The determinants of gender equity in India: Examining

Dyson and Moore's thesis with new data. *Population and Development Review*, *30*(2), 239–268. https://doi.org/10.1111/j.1728-4457.2004.012_1.x

Ridgeway, C. L., & Correll, S. J. (2004). Unpacking the gender system: A theoretical perspective on gender beliefs and social relations. *Gender & Society*, *18*(4), 510–531.

Risman, B. J. (2004). Gender as a social structure: Theory wrestling with activism. *Gender & Society*, *18*(4), 429–450.

Robinson, J. P., & Godbey, G. (1997). *Time for life: The surprising ways americans use their time*. Pennsylvania State University Press.

Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, *70*(1), 41–55. https://doi.org/10.1093/biomet/70.1.41

Sanchez, L., & Thomson, E. (1997). Becoming mothers and fathers: Parenthood, gender, and the division of labor. *Gender & Society*, *11*(6), 747–772.

Sarkisian, N., & Gerstel, N. (2016). Does singlehood isolate or integrate? Examining the link between marital status and ties to kin, friends, and neighbors. *Journal of Social and Personal Relationships*, *33*(3), 361–384. https://doi.org/10.1177/0265407515597564

Sayer, L. C. (2005). Gender, time and inequality: Trends in women's and men's paid work, unpaid work and free time. *Social Forces*, *84*(1), 285–303. https://doi.org/10.1353/sof.2005.0126

Sen, A. K. (1990). Gender and cooperative conflicts. In I. Tinker (Ed.), *Persistent inequalities: Women and world development* (pp. 123–149). Oxford University Press.

Shelton, B. A., & John, D. (1996). The division of household labor. *Annual Review of Sociology*, *22*, 299–322. https://doi.org/10.1146/annurev.soc.22.1.299

Solon, G., Haider, S. J., & Wooldridge, J. M. (2015). What are we weighting for? *Journal of Human Resources*, *50*(2), 301–316. https://doi.org/10.3368/jhr.50.2.301

South, S. J., & Spitze, G. (1994). Housework in marital and nonmarital households. *American Sociological Review*, *59*(3), 327–347. https://doi.org/10.2307/2095937

Umberson, D. (1992). Gender, marital status and the social control of health behavior. *Social Science & Medicine*, *34*(8), 907–917. https://doi.org/10.1016/0277-9536(92)90259-s

United Nations Statistics Division. (2016). *International classification of activities for time-use statistics 2016 (ICATUS 2016)*. United Nations.

Venn, S., Arber, S., Meadows, R., & Hislop, J. (2008). The fourth shift: Exploring the gendered nature of sleep disruption among couples with children. *British Journal of Sociology*, *59*(1), 79–97. https://doi.org/10.1111/j.1468-4446.2007.00183.x

Waite, L. J. (1995). Does marriage matter? *Demography*, *32*(4), 483–507. https://doi.org/10.2307/2061670

West, C., & Zimmerman, D. H. (1987). Doing gender. *Gender & Society*, *1*(2), 125–151. https://doi.org/10.1177/0891243287001002002